

Project Executable Files

Date	10/07/2026
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Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	3 Marks

Project Executable Files:

This section contains all executable and deployable files required to run the **Credit Card Approval Prediction** project. The project is built using Python and machine learning libraries to predict whether a credit card application should be approved or rejected based on applicant details. The solution includes the trained ML model, source code, datasets, and documentation for easy deployment and execution.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files	NA
5	Environment Configuration (.env if required)	No
6	Hosted / Deployed Application URL	No
7	APK / Executable Binary	NA
8	Dockerfile / Containerization config	No

9	Test files and test results	Yes
10	Demo Video / Walkthrough	Yes

Step 2: File / Folder Structure

VOICE-BASED-CONCEPT-UNDERSTANDING-ANALYSER/

```

|
|— app.py          # Main Streamlit application
|— requirements.txt # Project dependencies
|— README.md       # Project documentation
|— config.py       # Configuration settings
|
|— data/
|   |— reference_topics.json
|   |— sample_audio/
|
|— models/
|   |— whisper_model/
|   |— sentence_bert/
|
|— modules/
|   |— speech_to_text.py
|   |— semantic_analysis.py

```

```

|   ├── audio_analysis.py
|   ├── scoring.py
|   ├── report_generator.py
|   ├── utils.py
|
|   ├── assets/
|   |   ├── logo.png
|   |   ├── waveform.png
|   |
|   |   ├── reports/
|   |   |   ├── generated_reports/
|   |   |
|   |   ├── temp/
|   |   |   ├── uploaded_audio/
|   |   |
|   |   ├── screenshots/
|   |   |   ├── home_page.png
|   |   |   ├── result_page.png

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted
Login Credentials	Not Required
Platform / Hosting Provider	Local Machine (Python + Flask)

Repository Link	GitHub Repository (If Available)
Demo Video Link	Not Available

Step 4: Run Instructions

- Clone or download the VBCUA project repository.
- Open the project folder in VS Code.
- Create a virtual environment (optional).
- Install dependencies:
- `pip install -r requirements.txt`
- Run the Streamlit application:
- `streamlit run app.py`
- Open the browser.
- `http://localhost:8501`
- Upload an audio file (.wav or .mp3).
- Select the reference concept.
- Click **Analyze**.
- View:
- Speech Transcript
- Semantic Similarity Score

Step 5: Known Issues / Limitations

S.No	Known Issue	Workaround / Status
1	Works on local machine only	Can be deployed on Render, AWS, or Azure
2	Requires internet for Whisper model (if API used)	Offline Whisper models can be integrated
3	Limited predefined concepts	More concepts can be added easily